Geographic units, modern outcomes and sample filters

Controls: Select Analysis Level; Select Modern Year; Sector; Filter by Mining Share; Max Mining Share; Restrict to counties in an MSA.

1. Choose the observation. County analysis has one observation per county. MSA analysis has one observation per metropolitan statistical area, using the project's current county membership. State analysis has one observation per modern state and uses the CH model. An MSA can cross state borders. The selected modern year, from 2001 to 2022, determines when GDP and employment are measured; it does not select historical borders or the historical instrument year.

Let q_c, n_c and a_c denote county GDP, employment and land area. For an MSA or state g, write total GDP, employment and area as Q_g, N_g and A_g. The outcome is log GDP per employment:

$$y_c = \log(q_c/n_c), \qquad y_g = \log(Q_g/N_g).$$

Ordinary employment density is $d_c = n_c/a_c$ or $d_g = N_g/A_g$. Land area is measured in acres. The CH model instead uses the densities of the constituent counties.

2. Choose the sector for the outcome. County and MSA analyses offer All, Private, Manufacturing, Private non-farm, and Private non-farm/mining. Each sector outcome uses the corresponding GDP and employment. Private non-farm excludes agriculture, forestry, fishing and related activities; the final option also excludes mining. The density or employment regressor on the right-hand side continues to use total employment. Thus changing the sector changes whose productivity is measured, while retaining the same measure of surrounding economic activity. State estimates use all industries. Missing or suppressed components, or nonpositive GDP or employment, make the sector outcome unavailable.

3. Apply sample restrictions. The mining filter keeps an observation only when mining GDP divided by total GDP is known and no greater than the entered threshold. A threshold of 0.01 means one percent. An unknown mining share is excluded, rather than treated as zero. This filter is separate from removing mining from the sector outcome. It is not used in State analysis. Restrict to counties in an MSA removes nonmetropolitan counties from County analysis; it does not aggregate the remaining counties into MSAs.

4. Compare like with like. Changing the modern year, sector or filter can change the estimation sample. A coefficient difference may therefore reflect both a changed specification and changed observations. Check the observation count and regression configuration after running each model.